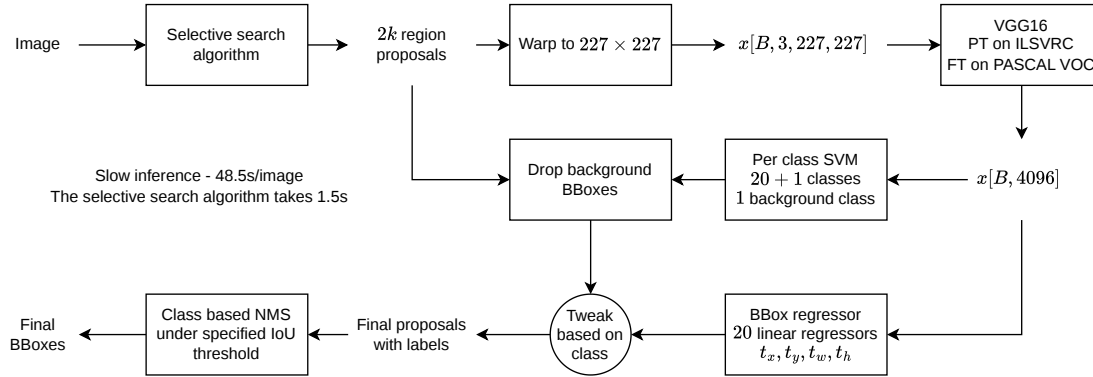


Practice Sheet 3

Q1. Draw an approximate flowchart of the RCNN framework.

Answer:



Note - It doesn't have to be this accurate.

Q2. Write down the limitations of RCNN framework that Fast RCNN tackled.

Answer:

1. RCNN was very slow in inference. It took 48.5 seconds to process one single image.
2. RCNN framework had a high space complexity. The SVMs and the linear regressors couldn't be processed end-to-end with the deep neural network. As such, all the embeddings had to be stored on disk before they can be passed to the prediction models.
3. For the same reason mentioned in the last point, RCNN cannot be trained end-to-end.

Q3. What is Non-Max Suppression? Why is it used?

Q4. Assume an object detection model outputs 4 bounding boxes for a certain class. The predicted confidence scores of the bounding boxes and the IoU values between them are provided as -

Bounding Box	Confidence	IoU	B ₁	B ₂	B ₃	B ₄
B ₁	0.70	B ₁	–	0.65	0.20	0.10
B ₂	0.85	B ₂	0.65	–	0.45	0.15
B ₃	0.90	B ₃	0.20	0.45	–	0.60
B ₄	0.60	B ₄	0.10	0.15	0.60	–

Find the resultant bounding boxes after performing NMS. Assume NMS IoU threshold to be 50%.

Answer: Sorting the BBoxes in descending order of confidence gives

$$\text{Candidate BBoxes} = [B_3, B_2, B_1, B_4]$$

We accept the highest confidence BBox, B_3 , and remove it from the candidate list:

$$\text{Accepted BBoxes} = [B_3]$$

$$\text{Candidate BBoxes} = [B_2, B_1, B_4]$$

We now compare the remaining bounding boxes with B_3 . Since the NMS IoU threshold is 0.5,

- $\text{IoU}(B_3, B_1) = 0.20 \leq 0.5 \rightarrow \text{retain } B_1$
- $\text{IoU}(B_3, B_2) = 0.45 \leq 0.5 \rightarrow \text{retain } B_2$
- $\text{IoU}(B_3, B_4) = 0.60 > 0.5 \rightarrow \text{suppress } B_4$

Thus,

$$\text{Candidate BBoxes} = [B_2, B_1]$$

We next accept B_2 and remove it from the candidate list:

$$\text{Accepted BBoxes} = [B_3, B_2]$$

$$\text{Candidate BBoxes} = [B_1]$$

The remaining candidate has

$$\text{IoU}(B_2, B_1) = 0.65 > 0.5,$$

so B_1 is suppressed.

Therefore, the resultant bounding boxes after NMS are $[B_3, B_2]$

Q5. Two bounding boxes predicted by an RCNN model for `class = cat` are $B_1 = [125, 120, 150, 120]$ and $B_2 = [155, 150, 150, 120]$. Calculate $\text{IoU}(B_1, B_2)$.

Note - The BBoxes follow center-width-height $[x_c, y_c, w, h]$ representation.

Answer: For B_1 ,

$$x_{min} = x_c - \frac{w}{2} = 125 - \frac{150}{2} = 50,$$

$$x_{max} = x_{min} + w = 200,$$

$$y_{min} = y_c - \frac{h}{2} = 120 - \frac{120}{2} = 60,$$

$$y_{max} = y_{min} + h = 180$$

For B_2 ,

$$x_{min} = x_c - \frac{w}{2} = 155 - \frac{150}{2} = 80,$$

$$x_{max} = x_{min} + w = 230,$$

$$y_{min} = y_c - \frac{h}{2} = 150 - \frac{120}{2} = 90,$$

$$y_{max} = y_{min} + h = 210$$

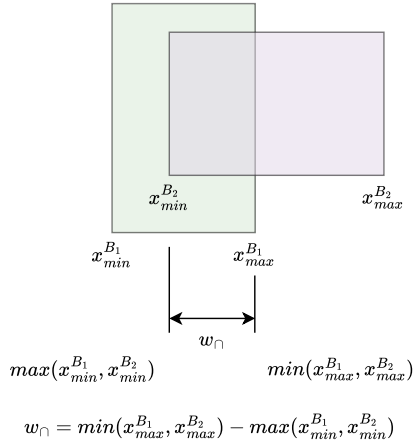
Now, we can find the overlap region's width and height,

$$\begin{aligned}
w_{\cap} &= \max\left(0, \min(x_{\max}^{B_1}, x_{\max}^{B_2}) - \max(x_{\min}^{B_1}, x_{\min}^{B_2})\right) \\
&= \max(0, \min(200, 230) - \max(50, 80)) \\
&= 120
\end{aligned}$$

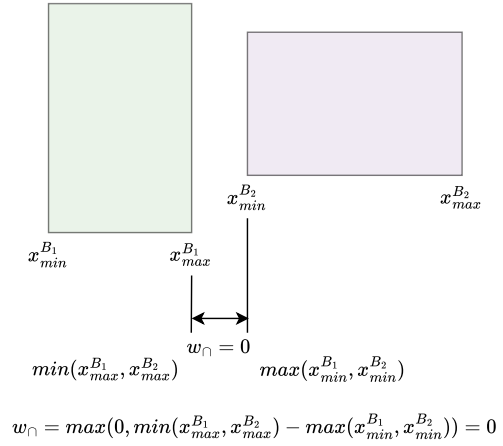
$$\begin{aligned}
h_{\cap} &= \max\left(0, \min(y_{\max}^{B_1}, y_{\max}^{B_2}) - \max(y_{\min}^{B_1}, y_{\min}^{B_2})\right) \\
&= \max(0, \min(180, 210) - \max(60, 90)) \\
&= 90
\end{aligned}$$

The complex formula is there to identify zero overlap. You can approach it geometrically too.

For overlapping bboxes



For non-overlapping bboxes



Using the overlap region's width and height,

$$\text{area}(B_1 \cap B_2) = w_{\cap} \cdot h_{\cap} = 120 \cdot 90 = 10800$$

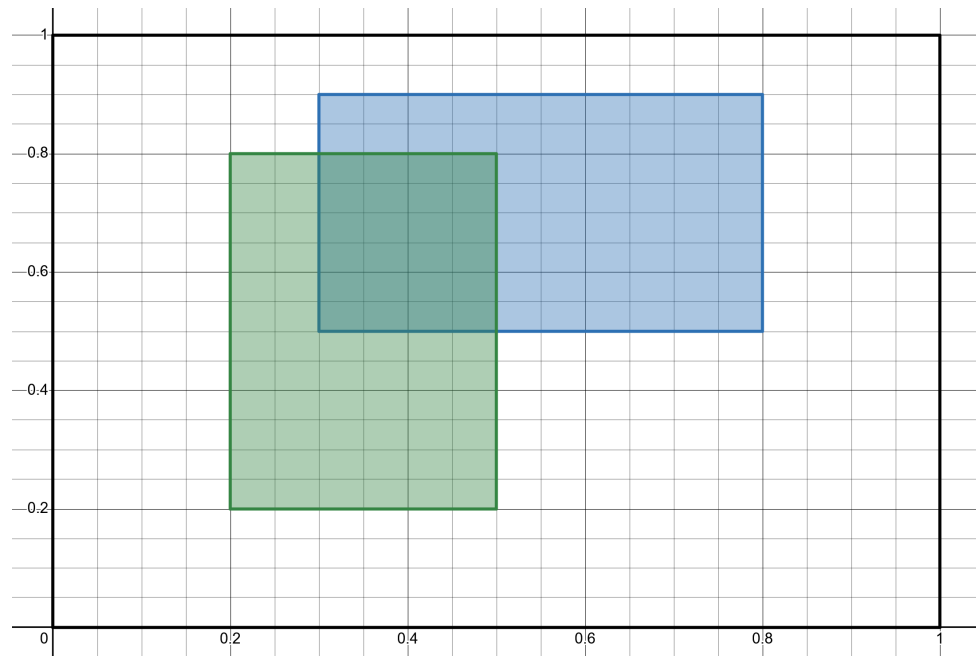
The area of union is

$$\begin{aligned}
\text{area}(B_1 \cup B_2) &= w^{B_1} h^{B_1} + w^{B_2} h^{B_2} - \text{area}(B_1 \cap B_2) \\
&= 150 \cdot 120 + 150 \cdot 120 - 10800 \\
&= 18000
\end{aligned}$$

Therefore,

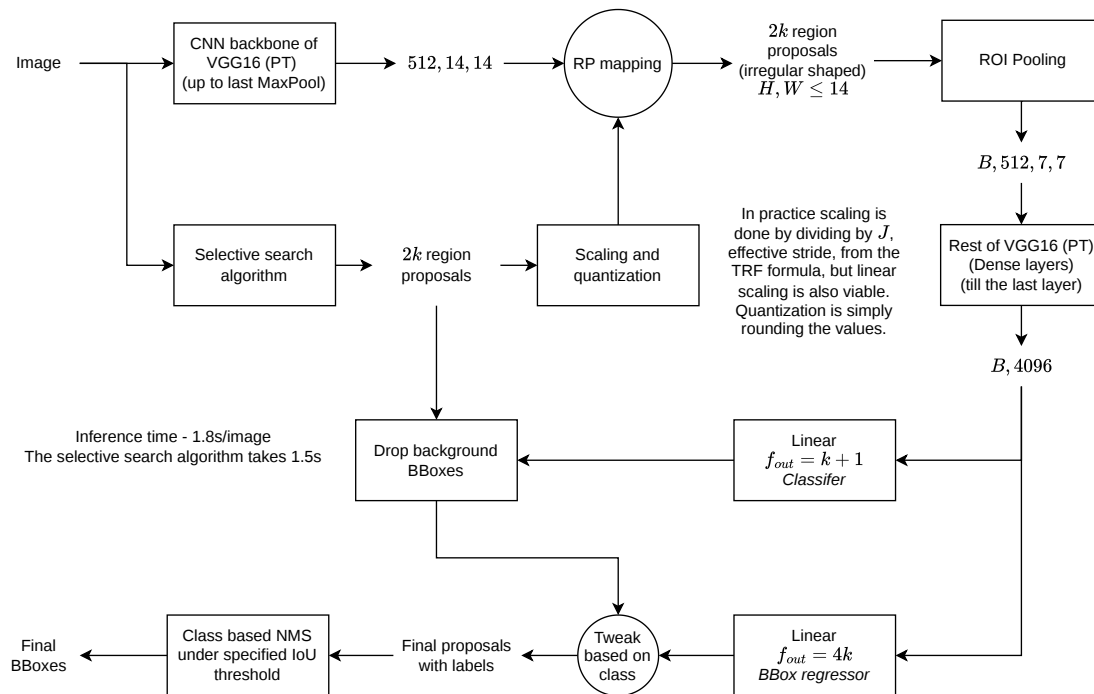
$$\text{IoU}(B_1, B_2) = \frac{\text{area}(B_1 \cap B_2)}{\text{area}(B_1 \cup B_2)} = \frac{10800}{18000} = 0.6.$$

Q6. Find the IoU of between the two colored regions.



Q7. Draw an approximate flowchart of the Fast RCNN framework.

Answer:



Note - It doesn't have to be this accurate.

$k = 20$ for PASCAL VOC

Q8. What is the task of ROI Pooling in Fast RCNN?

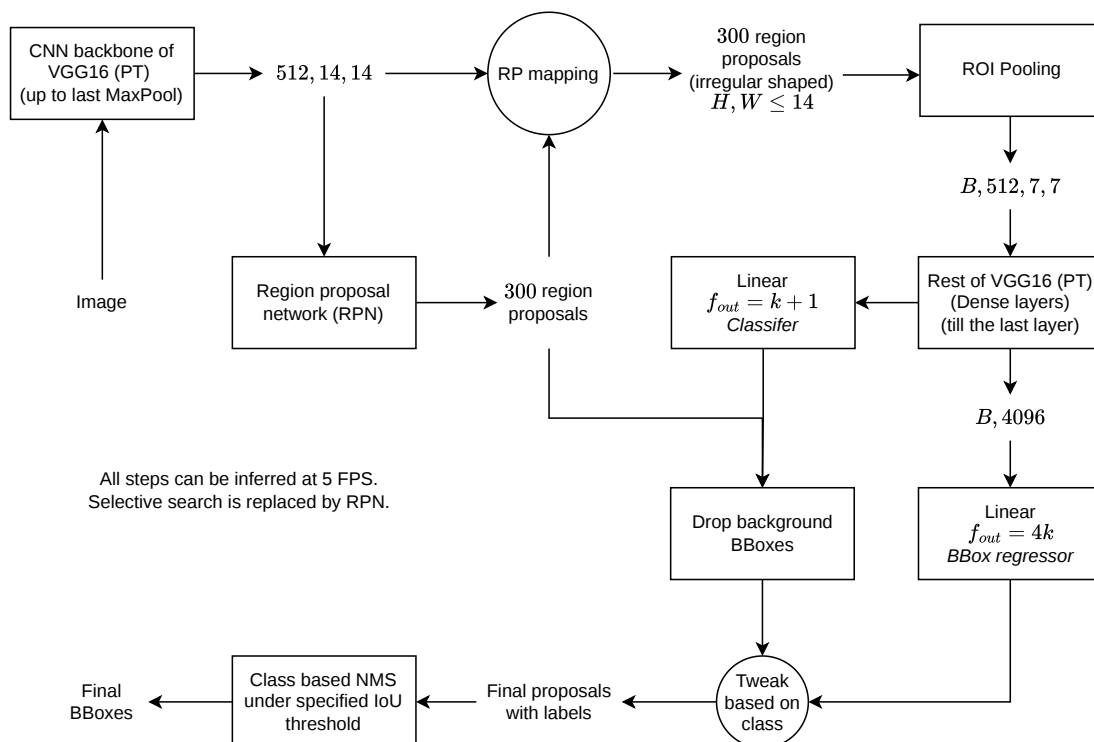
Answer: ROI Pooling converts the irregularly shaped region proposals to uniform 7×7 feature maps.

Q9. What was the main bottleneck of Fast RCNN in terms of inference speed?

Answer: Selective search algorithm was the main performance bottleneck of Fast RCNN, a problem that Faster RCNN stops.

Q10. Draw an approximate flowchart of Faster RCNN.

Answer:



Q9. Bangladesh government wants to place AI cameras to detect the number plates of cars that violate the traffic laws at key junctions of Dhaka. To process captured images for OCR, the ICT ministry opted for a convolutional backbone. Considering the system needs to perform on-chip, which SotA model can be used as a backbone to the system?

Q10. Ihzaz developed two models, `model_A` having 10M parameters and `model_B` having 50M parameters. However, during inference, `model_B` was found to be faster. Why may this happen?